

Topic 1.7 - Rational Functions and End Behavior

Guided Notes

Strengthened ECHS edition - reasoning, challenge, and AP-style practice

Name: _____

Class: _____

Date: _____

Why this topic matters

Rational functions often describe saturation, average cost, efficiency, and ratios. Their long-run behavior is not always a horizontal line: polynomial division can expose a slant or higher-degree asymptote, and the remainder describes the shrinking deviation from that model.

Learning targets

- Determine rational end behavior by comparing numerator and denominator degrees.
- Use polynomial division to identify slant or polynomial asymptotes.
- Analyze approach from above or below and interpret long-run behavior in context.

Language and structure

Term	Meaning in this topic
horizontal asymptote	A constant long-run model $y = L$.
slant asymptote	A linear long-run model found by division when degree difference is 1.
polynomial asymptote	The quotient polynomial when numerator degree exceeds denominator degree.
proper rational function	A rational expression with numerator degree below denominator degree.
remainder form	$R(x) = Q(x) + \frac{r(x)}{d(x)}$, making deviation from Q visible.

Launch - reason before calculating

Compare the long-run behavior of $A(x) = \frac{3x^2-5}{x^2+4}$ and $B(x) = \frac{3x^2-5}{x^3+4}$. Explain why the same numerator does not produce the same asymptote.

Concept 1: Degree comparison

Core idea

If numerator degree is less than denominator degree, the horizontal asymptote is $y = 0$. If degrees are equal, it is the ratio of leading coefficients. If numerator degree is greater, division is needed; there is no horizontal asymptote from a leading-coefficient ratio.

Worked example - annotate the reasoning

Problem. Find the asymptote of $R(x) = \frac{3x^2-5}{x^2+4}$ and determine whether the graph approaches from above or below.

Model reasoning. Asymptote $y = 3$. Since $R - 3 = \frac{-17}{x^2+4} < 0$, the graph approaches from below on both ends.

Check your understanding

Independent

No calculator

Find the horizontal asymptote of $S(x) = \frac{-4x^3+x}{2x^3-7}$.

Concept 2: Polynomial division and slant asymptotes

Core idea

When numerator degree is at least denominator degree, divide. The quotient is the polynomial asymptote because the proper remainder term tends to zero when its numerator has lower degree than the denominator.

Worked example - annotate the reasoning

Problem. Rewrite $\frac{2x^2+3x-1}{x-2}$ in quotient-remainder form.

Model reasoning. Division gives $2x + 7 + \frac{13}{x-2}$, so the slant asymptote is $y = 2x + 7$.

Check your understanding

Independent

No calculator

Find the polynomial asymptote of $\frac{x^3-2x+5}{x^2+1}$.

Concept 3: Approach and contextual meaning

Core idea

The sign of $R(x) - Q(x)$ shows whether the graph lies above or below its asymptote. In context, the asymptote may represent limiting cost, concentration, capacity, or a trend line. A horizontal asymptote can be crossed; it is not a domain restriction.

Worked example - annotate the reasoning

Problem. For $F(x) = 5 + \frac{2x}{x^2+1}$, describe approach to $y = 5$ on both ends.

Model reasoning. For large positive x , the correction is positive, so approach from above. For large negative x , it is negative, so approach from below.

Check your understanding

Independent

No calculator

For $G(x) = 8 - \frac{3}{x^2+4}$, describe the approach.

Representation laboratory

Connect at least two representations

A graph follows the line $y = -2x + 3$ more closely as $|x|$ grows. Write two different rational functions with that slant asymptote, one above the line for large positive x and one below it.

Challenge laboratory

Challenge - justify every claim

Construct a rational function with horizontal asymptote $y = -2$, approached from above as $x \rightarrow \infty$ and from below as $x \rightarrow -\infty$, with no real vertical asymptotes.

Error clinic and calculator strategy

Common errors to diagnose

- Using leading-coefficient ratio when degrees are unequal.
- Stopping division before the remainder degree is below the divisor degree.
- Treating an asymptote as a line the graph can never cross.
- Ignoring the sign of the remainder when describing approach.

Technology decision

Graph the rational function together with its asymptote and graph their difference. A wide window can show long-run agreement, while a difference table reveals approach direction. Algebraic division remains the most reliable way to identify the exact asymptote.

AP-style synthesis**Multi-part reasoning task**

Let $R(x) = \frac{2x^3 - x^2 + 5}{x^2 - 4}$.

- A. Use division to write R in quotient-remainder form.

- B. State the polynomial asymptote and excluded inputs.

- C. Determine whether R is above or below its asymptote for sufficiently large positive x .

Exit ticket**Before you leave**

1. State the three degree-comparison cases.

2. What part of division gives the polynomial asymptote?

3. Can a rational graph cross a horizontal asymptote?
